Comparison of Web size measures for predicting Web design and authoring effort

E. Mendes, N. Mosley and S. Counsel!

Abstract: Software practitioners recognise the importance of realistic estimates of effort for the successful management of software projects, the Web being no exception. Estimates are necessary throughout the whole development lite cycle. They arc fundamental when bidding for a contract or when determining a project's reasibility in terms of cost-benefit analysis. In addition, they allow project managers and development organisations to manage resources effectively. Size, which can be described in terms of length, functionality and complexity, is often a major determinant of effort. Most effort prediction models to date concentrate on functional measures of size. although length and complexity arc also essential aspects of size. A case study evaluation is described, in which size metrics characterising length, complexity and functionality arc obtained and used to generate etTort prediction models for Web authoring and design. The comparison of these size metrics as effort predictors is described by generating corresponding prediction models, and their accuracy is compared using box plots of the residuals. Results suggest that in general all categories present a similar prediction accuracy.

1 Introduction

Software practitioners recognise the importance of realistic estimates or effort !or the successful management of software projects. the Web being no exception. Prediction is a necessary part of an effective process [I], be it authoring, design, testing or Web development as a whole.

This paper focuses on effort prediction for design and authoring processes. We adopt the classification proposed by Lowe and Hall [2], where authoring represents the creation or content, structure and presentation of the application, while design covers the methods used for generating the structure and functionality or the application.

The data used to generate our prediction models was gathered through a quantitative case study evaluation, where a set or proposed or reused [3-6] size metrics for effort prediction were measured. The prediction models proposed were generated by statistical techniques, specifically linear and stepwise regression.

2 Related work in Web engineering

There are relatively few examples of effort prediction models for Web development in the literature. and most propose methods and tools as a basis for process improvement and higher product quality [7-10].

Morisio *et al.* [I lj describe an experiment involving the development of Web-based applications, using an object-oriented framework, where function and code size metrics were collected. They apply an existing effort prediction model based on different reuse types and conclude that classical function points arc not appropriate for the environment they use.

Rollo [12] explores the issues of counting Web sites using IFPUG [13] and MKII [14] methods, discussing difficulties and contradictions between the rules and approaches to Web site sizing. In addition, the same Web site is counted using the sizing method COSMIC-HP [Note I] [15]. He concludes that COSMIC is the most flexible approach for counting the functional size of any Web site.

Mendes *et al.* [16] describe a case study where size and effort for Web applications, structured according to the cognitive flexibility [17], are measured and used to generate prediction models (linear and stepwise regression and analogy), which are then compared. Results show that the best prediction is obtained using analogy.

The contribution of the above to the refinement of our case study is as follows: ti) consideration of reuse is seen to be fundamental when sizing Web applications, (ii) COSMIC-FFP presented good results, which needed to he validated on other datasets, (iii) confirmation is provided that size measures are predictors of effort, and (iv) the need for objective size measures collected automatically whenever possible is evident.

3 Collecting size metrics for Web authoring and design effort prediction

3.1 Introduction

The case study evaluation measured the size of Web applications and their design and authoring effort. Four metrics· representing confounding factors were also measured.

Our measurement activity was developed according to Fenton and Pfleeger's conceptual framework for software measurement [3], which they describe using three principles:

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E. Mendes is with the Department of Computer Science, University of Auckland, Private Bag 92019, Auckland, New Zealand

N. Mosley is with MxM Technology, PO Box 3139, Shortland St., Auckland, New Zealand

S. Counsel! is with the Department of Computer Science, Birkbeck College, University of London, Senate House, Russell Square, London, UK

Note 1: COSMIC-FFP (common software measurement international consortium - full function points).

Table 1: Classification of products, processes and resources for case study

Entities	Attributes (internal)
Products	
Web application	page count media count program count total page allocation total media allocation total embedded code length reused media count reused program count total reused media allocation total reused code length connectivity connectivity density total page complexity cyclomatic complexity sizecFsu structure
Processes	
Web authoring and design processes	total effort
Resources	
Developer	authoring and design experience
Authoring tool	tool type

l. Classi:(ying the entities to be examined.
2. Determining relevant measurement goals.
3. Identifying the level of maturity the organisation has reached.

The classification of entities applied to the case study is presented in Table I.

All metrics presented here arc described in detail in Section 3.3.

Our measurement goals were documented using the goal-question-metric (GQM) [18] (Table *2).*

Finally. the level of maturity within the Web application development community considered for our case study is L measured according to the capability maturity model (CMM) [19].

3.2 Case study

The case study consisted of the design and aLtlhoring of Web applications aimed at teaching a chosen topic, structured according to the cognitive flexibility theory (CFT) principles [17], using a minimum of 50 Web pages, and representing a medium-size Web application [2]. All subjects received training on the CFT authoring principles for approximately 150 min.

The relationship between the process model [2] used by the students to develop the applications and the steps to apply the CFT are presented in Fig. 1.

Forty-three computer science students participated in the case study. Two questionnaires were used to collect the data. The first [Note 2] asked suhjccts to rate their Web authoring experience using an ordinal scale, from no experience (one) to very good experi<:ncc (five). The second questionnaire (Note 3] \Vas used to measure size, confounding factors and design and authoring effort. On both questionnaires. each scale type was detailed to avoid misunderstandings.

To reduce learning effects, subjects were given a coursework prior to designing and authoring the Web applications. consisting of the creation of a Web site for the Matakohc Kauri Museum [Note 4], improving on their existing site.

Early analysis of the data showed that four questionnaires were missing fundamental information. They were not used in the data analysis. In addition, we identified

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Table 2: Goals, questions and metrics of case study

Goal	Question	Metric
Purpose: to measure	What attributes can characterise size of Web applications?	page count media count
Issue: Web design and authoring processes		program count total page allocation total media allocation
Object: process		total embedded code length reused media count
Viewpoint: developer's		reused program count viewpoint total reused media allocation total reused code length connectivity connectivity density total page complexity cyclomatic complexity sizecFsu structure
	What influence can developers have on effort to design and author Web applications?	authoring and design experience
	How can design and authoring processes be measured?	total effort
	What influence can authoring tool have on effort required to author Web application?	tool type

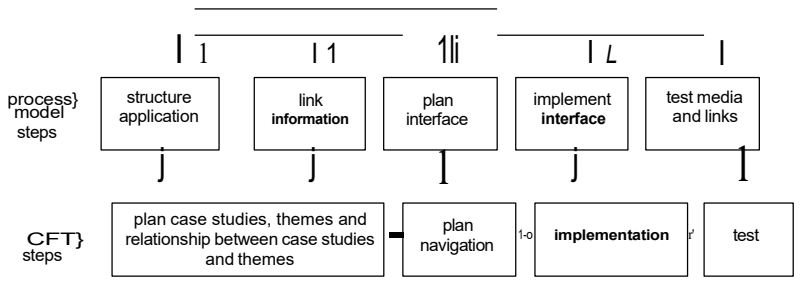

Fig. 1 *Relationship between process model and CFT steps*

Table 3: Length metrics

Metric	Description
Page count*	number of html or shtml files used in Web application
Media count*	number of non-reused media files used in application
Program count	number of non-reused cgi scripts (21], Javascript [22] files, Java applets [23] used in application
Total page allocation*	total space (IVIbits) allocated for all html or shtml pages used in application
Total media allocation*	total space (Mbits) allocated for all media files used in application
Total embedded code length	total number of lines of code for all programs used by application
Reused media count*	number of reused/modified media files
Reused program count	number of reused/modified programs
Total reused media allocation"	total space (Mbits) allocated for all reused media files used in application
Total reused code length	total number of lines of code for all programs reused by application

• Suitable for static Web applications where number of dynamically generated links and/or pages is absent

Table 4: Complexity metrics

Metric	Description
Connectivity*	Total number of internal linkst (not including dynamically generated links)
Connectivity density 131*	connectivity/page count
Total page complexity*	I:fage *count* no. of different types of media in page/page count
Cyclomatic complexity [31*	(connectivity - page count)+ 2

* Suitable for static Web applications where number of dynamically generated links and/or pages is absent
t Subjects did not use external links to other Web sites; all links pointed to pages within application only

Table 5: Functionality metric

Metric	Description
SizecFsu [15]	I.: size (entries)+ I: size (exits) , I_: size (reads)+ I.: size (writes)

Table 6: Effort metric

Metric	Description
Total effort	estimated elapsed time (number of hours) it took subject to design and author application (summation of elapsed times spent on each process model step)

three datapoints in the dataset for which the total effort to develop an application was noticeably high, two of which did not present reasonable justification in the data (applications were small in size and experience was not low). These were therefore removed from the dataset, leaving 37 datapoints.

3.3 *Metrics collected during case study*

The metrics collected during the case study evaluation represented attributes of three categories: Web application size, design and authoring effort, and confounding factors [20]. Size metrics were organised in three tables (Tables 3-5) according to the categories of length, complexity and functionality. Effort, measured in person/hours, is presented in Table 6. Finally, confounding factors which could affect the internal validity of the experiment are presented in Table 7.

Media and program files are considered reused only if they already existed externally to the Web application, and are copied for use (as a black box or not) within the application.

Compactness and stratum [25] were not considered in this evaluation as they could only be measured subjectively, compromising the validity of the results.

3.4 *Measuring* size *metrics*

Complexity and length metrics were collected using a questionnaire and three spreadsheets which subjects were required to complete. All complexity and length metrics, except reuse, were re-measured by using the applications developed.

functionality was measured using the COSMlC-ffP [15) measurement method. We counted the number of cosmic functional size units (CFSUs) in the Web applica-

Table 7: Confounding factors

Metric	Description
Tool type	measures type of tool used to author/design Web pages: WYSIWYG. (what you see is what you get), semi-WYSIWYG or text-based
Authoring and design experience	measures authoring/design experience of subject
Structure*t	measures how main structure (backbone) of application is organised

*Suitable for static Web applications where number of dynamically generated links and/or pages is absent
tThe structure can be a sequence, hierarchy or network. A sequential structure corresponds to documents linearly linked, a hierarchical structure denotes documents linked in a tree shape, and a network structure denotes documents linked in a net shape [24l

tion. The COSMIC-FFP method is a standardised measure of software. It is applicable to domains such as application software (banking, insurance, accounting, etc.), real-time software, and hybrids of both. We have adapted the method to Web applications in a way similar to [12].

The COSMIC-FFP method involves applying a set of rules and procedures to a given piece of software, as perceived from the perspective of its functional user requirements (FURs) [Note S]. The result is a numerical 'value of quantity' [Note 6] representing the functional size of the software. The model consists of two parts: the context model and the software model.

The context model helps establish what is part of the software and what is part of the software's operating environment. It identifies the software boundaries and the functional flow of data attributes. This flow has two directions, front-end and back-end, representing, respectively, entries and exits and reads and writes (see Fig. 2).

The software model assumes that the software to be mapped and measured manipulates pieces of information designated as data groups, which consist of data attributes. The software functional requirements are implemented -by a set of functional processes, each of which is an ordered set of sub-processes performing either data movement or data manipulation (see Fig. 3).

The functional size of software is directly proportional to the number of its data movement sub-processes. The unit of measure used is 1 CFSU, equivalent to a single data movement at the sub-process level. Consequently, the total size is given by the number of entries+ number of exits+number ofreads+number of writes [15].

The context model for Web applications in the case study is presented in Fig. 4.

The counting rules used corresponding to each entry, exit, read and write are as follows:

• Each 'HREF' tag counted as I entry+ 1 read+ 1 exit. By pressing a link, the user sends an entry to the application, which reads the data from the Web server and shows the contents to the user. We assumed the Web server to be the data storage hardware. Although several subjects used

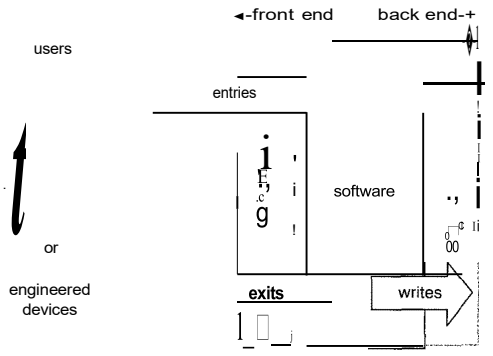

Fig. 2 *Genetic flow of data attributes,from functional perspective [15}*

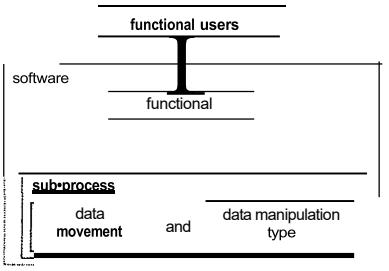

Fig. 3 *General software model for measuring fimctional size [15]*

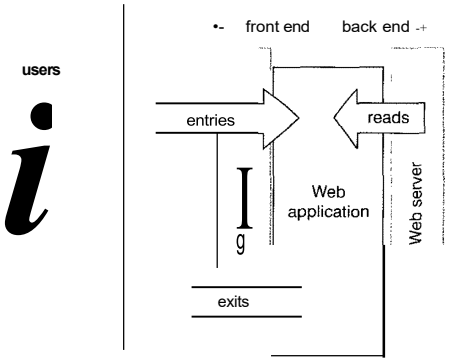

Fig. 4 *Flow of data attributes through Web application from functional perspective*

in-line images [21], they were not considered as a separate data group since every 'IMG SRC' tag will be downloaded automatically with the Web page. The subjects who participated in this case study did not use external resources (images, sound and video), which use the 'HREF' tag and have to be downloaded from the Web server upon request.

• In relation to the types of programs used, subjects used either Java applets or Javascript. For each Java applet we counted one entry+ one exit (one entry to activate the applet and one exit showing the applet). As applets are also downloaded from the server at the same time as the Web page, we did not consider a 'read' flow of data.

• For each Javascript file, we only considered the 'HREF' tag ('entry' flow of data) to be the source of interaction between user and application.

Note 5: The term 'functional user requirement' is an ISO expression designating a subset of the user requirements. It represents the user practices and procedures that the software must perform to fulfil the user's needs, excluding quality requirements and any technical requirements [1S0/IEC 14143:1997].
Note 6: The value of a quantity is the magnitude of that particular quantity, generally expressed as a unit of measurement multiplied by a number [IOS-1993].

We did not have a functional users requirements document for the Web applications, so we counted the number of CFSUs for each application using the implementation provided by the subjects only. Some links ('HREF' tags) existed to give readers more information about the contents of the application; others to give better navigation guidance (site map, back to main page, previous page, next page, etc.). We understand that both types of links are equally important in achieving the purpose of conveying information in a meaningful way with a reasonable level of functionality. We believe that our counting methodology would map easily to any document in which navigation and linking topologies were represented.

3.5 Validity of case study

Comments on the validity of the case study are as follows:

• There are difficulties measuring Web applications because: (i) design and authoring effort are difficult concepts to measure, (ii) there is no standard as to what constitutes the design and authoring processes of a Web application. Therefore, the activities used in this study may not be representative of all practices employed in the Web community.
• The metrics collected, apart from effort, experience and structure, were all objective and precisely quantifiable. The granularity of effort data was made consistent with the Web authoring process employed. In addition, the scale used to measure experience and the three types of structure of an application were described in detail in both questionnaires.
• There were four confounding factors in the case study evaluation:

(i) The subjects' authoring and designing experience.
(ii) Maturation effects, i.e. learning effects caused by subjects learning as an experiment proceeds.
(iii) The structure of the application.
(iv) Tools used to help author and design the Web application.

In relation to those confounding factors, the data collected revealed that:

(i) Subjects' authoring and design experiences were mostly scaled little (experience= 2) or average (experience= 3), with a low skill differential [Note 7]. A scatter plot between experience and effort was used to investigate their relationship. Most datapoints fell within clusters between the intervals 10 to 80 h, either for experience 2 or 3. Consequently, the original dataset was left intact.
(ii) Subjects had to develop a Web application as part of their first assignment. They also received training in the CFT principles, reducing maturation effects.
(iii) Notepad (or a similar text editor) and FirstPage were the two tools most frequently used. Although they differ with respect to the functionality offered, a scatterplot between total effort and tool revealed that for both tools most datapoints fell within the interval between 10 to 80 h. Consequently, confounding effects from the tools were reduced.

• The selection effects, which are due to natural variations in subject performance, were reduced as individual differences were spread across all applications.
• The instrumentation effects in general did not occur in this evaluation; the questionnaires used were the same.
• The results may he domain dependent, as all subjects answered the questionnaires based on their experience in developing Web applications for education. This evaluation should therefore be repeated in domains other than education if the results are to be generalised to other domains.
• The applications did not have more than 103 documents (mean=57) and 1501 links (mean 616). These numbers represent medium size applications, according to [2]. Consequently, they may not be representative of the large Web applications for education developed by some organisations. However, the applications developed had interface and contents quality similar to or better than Web applications developed by professionals (as observed by one of the authors when marking the applications), suggesting that the results of the case study are likely to scale-up.
• A threat to subject generalisability [26] may exist when the subject population is not drawn from a wider population. The subjects that participated in the case study were either final-year undergraduate students or MSc students. It is likely that they present skill sets similar to Web professionals at the start of their careers. The use of students as subjects, while sometimes considered unrealistic, is justified for two reasons: firstly, empirical evidence by Boehm-Davis and Ross [27] indicates that students are equal to professionals in many quantifiable measures, including their approach to developing software; secondly, for pragmatic considerations, having students as subjects was the only viable option for this case study.

4 Estimation process

4.1 Introduction

For the collected data, both linear and stepwise regressions [28, 29] were used to generate effort prediction models for Web design and authoring, which were then compared according to their prediction accuracy. Stepwise regression has been frequently used as a benchmark [29-33], and is regarded by some as a good prediction technique [34).

When applying linear regression, whenever we had more than one predictor variable to generate a model, we used all variables simultaneously.

Both linear regression and stepwise regression were calculated using SPSS 10.0.5.

4.2 Results using linear and stepwise regression

The predictors and response variable had a linear relationship. Before generating the prediction models we looked for a statistically significant correlation between predictors in order to avoid spurious relationships. In addition, we generated plots of the residuals, which revealed distributions that did not present any disturbing patterns.

The formulas obtained by using linear regression and stepwise regressions are shown in Table 8.

The metrics used to generate the prediction models were measured on a ratio scale.

In relation to length. both prediction models included page count and total reused code length. We did not assume that either page count or total page allocation would have a spurious relationship with efl<.1rt since they did not have a statistically significant correlation, and in practice it is quite possible to have Web applications with a small number of pages and using a good amount of storage space. The opposite is likely to happen as well, in particular when the application is more text-based than graphics-based.

In relation to complexity, we only used connectivity and total page complexity in the regression models, as there were strong statistically significant correlations between

i\\)te 7: **Lo\v** difference between skill levels.

Table 8: Prediction models using linear and stepwise regression

Size metrics	Linear regression	Adjusted $R2$	Stepwise regression	Adjusted $R2$
Length	-3.158+0.558* page count+ 0.03937* total page allocation+ 0.02795* total reused code length	0.421	2.963+0.03374* total reused code length+ 0.506 * page count	0.418
Complexity	20.911 + 0.280 * connectivity	0.310	17.840+0.02703* connectivity	0.355
Functionality	20.061 + 0.00895* sizecFsu	0.378	20.061 +0.00895* sizecFsu	0.378

connectivity and connectivity density, and connectivity and cyclomatic complexity. Connectivity presented the best statistical correlation with effort, when compared to connectivity density and cyclomatic complexity. As shown in Table 7, both regression models for complexity included connectivity only.

In relation to functionality, both prediction models included sizecFSU•

Unfortunately, no model presented reasonable prediction based on their adjusted R_2, indicating that replications of our case study are clearly necessary to investigate if possible bias in the manual data collection influenced our results.

4.3 Comparing prediction models

We used boxplots of the residuals to compare the two prediction models generated, for each one of the size categories, as suggested by Pickard *et al.* [35] and Kitchenham *et al.* [36]. It gives a good indication of the distribution of the residuals and can help explain the behaviour of the summary statistics [37].

The boxplots of the residuals (difference between the actual and predicted values for effort), presented in Fig. 5, were used to compare:

- The range of the residuals from each of the different models, for each size category.
- The distribution of the residuals.

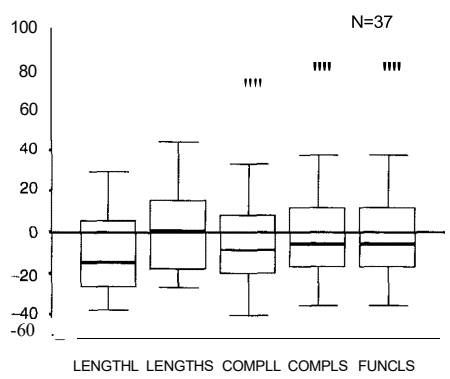

Fig. 5 *Boxplot of residuals for each prediction system used, organised by size category*

LENGTHL: residuals for length based on estimated effort generated using linear regression model •
LENGTHS: residuals for length based on estimated effort generated using stepwise regression model
COMPLL: residuals for complexity based on estimated effort generated using linear regression model
COMPLS: residuals for complexity based on estimated effort generated using stepwise regression model
FUNCLS: residuals for functionality based on estimated effort generated using linear and stepwise regression models (as both revealed same results, only one box used)

- The number of outliers.

The boxplots obtained indicate that:

- The distributions of COMPLL, COMPLS and FUNCLS have a slight right skew.
- The distributions for LENGTHL and LENGTHS are positively and negatively skewed, respectively.
- COMPLL, COMPLS and FUNCLS presented one outlier each, which is the same datapoint. Removing the outlier from the dataset did not influence the equations produced.
- LENGTHL and LENGTHS have a slightly tighter spread than COMPLL, COMPLS and FUNCLS, being less variable and less peaked.
- Except for LENGTHS, all medians represented values below zero, meaning that the estimates were biased towards over-estimation.
- For LENGTHL, COMPLL, COMPLS and FUNCLS, more than 50% of datapoints are negative (below zero), indicating estimated effort greater than actual effort (over-estimation).
- All distributions seem to be flatter as the size of the boxes are not small when compared to the tails.
- None of the models produced reasonable accurate estimates of effort.

We also tested the statistical significance of the absolute residuals for all size categories to test whether the differences exist by chance alone or are legitimate.

The statistical significance of the results was tested using a T-test of mean difference between paired residuals and a Mann-Whitney U-test, similar to [38]. However, as the data used is skewed, preference should be given to the results presented using the Mann-Whitney U-test, as it is a non-parametric test. For the results obtained, see Table 9.

No statistical significance was found, indicating that the models did not produce significantly different residual values. Consequently, we can conclude that there was no single technique that produced a better fitting model than the others.

The case study's goal was achieved, although the manual collection of part of the data, in particular effort, may have influenced the results obtained.

5 Conclusions and° future work

The first half of this paper presented a case study evaluation, in which we measured attributes of Web applications corresponding to three size categories, namely length, complexity and functionality. The second half of the paper suggested effort prediction models generated for each one of the size categories, using two statistical techniques, linear regression and stepwise multiple regression. These models were compared using boxplots of the residuals, and none presented an accurate estimate of effort. In addition, there was no statistical significance of

Table 9: Statistical significance for absolute residuals

T-test						Mann-Whitney U-test					
Linear regression			Stepwise regression			Linear regression			Stepwise regression		
AR1	**AR2**	SF?	AR1	AR2	SF?	**AR1**	AR2	SF?	AR1	AR2	SF?
length	com	N	length	compl.	**N**	length	compl.	N	length	compl.	N
length	function.	N	length	function.	**N**	length	function.	N	length	function.	**N**
compl.	function.	**N**	compl.	function.	**N**	compl.	function.	**N**	compl.	function.	**N**

ARI =absolute residual 1, AR2=absolute residual 2, SF?=statistically significant for a=0.05?, compl.=complexity, function.=functionality

the absolute residuals, and so no single technique to produce a better fitting model than the others.

At least for the Web applications used in our dataset, it seems that measuring the functionality of those applications was very much related to the number of links that the application has. As we counted the number of cosmic functional size units using the final implementation, rather than the functional user requirement model, further investigation is clearly necessary to determine whether or not functionality is highly correlated to connectivity.

Our future work includes using the COSMIC-FFP model to measure the functionality of several types of static and dynamic Web sites, and to use real datasets from industrial practice.

To conclude, there is an urgent need for adequate Web development effort prediction at an early stage in the development. As the use of the Web as a resource delivery environment increases, effort estimation can contribute significantly to the reduction of costs and time involved in developing Web applications.

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